

# **JNTU Hyderabad**

## **MODEL QUESTION PAPER (R20 PATTERN)**

Model Question Paper 2 ? Operating Systems (Latest Pattern)

Subject Code: CS402 Semester: 4 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Process Management)**

1. Explain process states and the Process Control Block (PCB) with a neat state transition diagram. [12 Marks]
2. Differentiate between process and thread. Explain multithreading models with examples. [12 Marks]
3. How is Process Management implemented? Give an algorithm / procedure and explain. [12 Marks]

### **SECTION B (Unit 2: CPU Scheduling)**

4. Explain CPU scheduling algorithms ? FCFS, SJF, Round Robin and Priority ? with suitable examples and Gantt charts. [12 Marks]
5. Calculate the average waiting time and average turnaround time for the given set of processes using Round Robin scheduling (time quantum = 2). [12 Marks]
6. How is CPU Scheduling implemented? Give an algorithm / procedure and explain. [12 Marks]

### **SECTION C (Unit 3: Memory Management)**

7. Explain paging and segmentation memory management techniques. Compare paging and segmentation. [12 Marks]
8. What is virtual memory? Explain the working of demand paging with necessary diagrams. [12 Marks]
9. Solve a problem related to Memory Management. Show all the steps clearly. [12 Marks]

### **SECTION D (Unit 4: File Systems)**

10. Explain file system architecture, directory structure and allocation methods ? contiguous, linked and indexed. [12 Marks]
11. Define File Systems. Explain its significance in Operating Systems with a suitable example. [12 Marks]
12. Solve a problem related to File Systems. Show all the steps clearly. [12 Marks]

### **SECTION E (Unit 5: Deadlocks and Synchronization)**

13. What is a deadlock? Explain the necessary conditions for deadlock and the methods

**for handling it. [12 Marks]**

14. Explain the Banker's algorithm for deadlock avoidance with an example. How is the safe state determined? [12 Marks]

15. Explain the producer?consumer problem and the readers?writers problem with semaphore solutions. [12 Marks]

\*\*\* End of Question Paper \*\*\*